Time Series Analysis and Forecasting of Coffee Prices

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```
##parta
#download the data
library(quantmod)
symbol <- "KC=F"
start_date <- "2025-01-01"
end_date <- "2026-01-01"
getSymbols(symbol, src="yahoo", from=start_date, to=end_date)

## [1] "KC=F"

head(`KC=F`)

##            KC=F.Open KC=F.High KC=F.Low KC=F.Close KC=F.Volume
KC=F.Adjusted
## 2025-01-02    321.55    329.80   319.75     326.85       14736
326.85
## 2025-01-03    324.75    326.25   317.65     318.65       11892
318.65
## 2025-01-06    318.65    328.60   317.05     318.60       15989
318.60
## 2025-01-07    320.65    323.10   317.75     320.50       12720
320.50
## 2025-01-08    320.65    324.30   314.90     316.45       21114
316.45
## 2025-01-09    321.00    322.00   316.65     318.50       14253
318.50

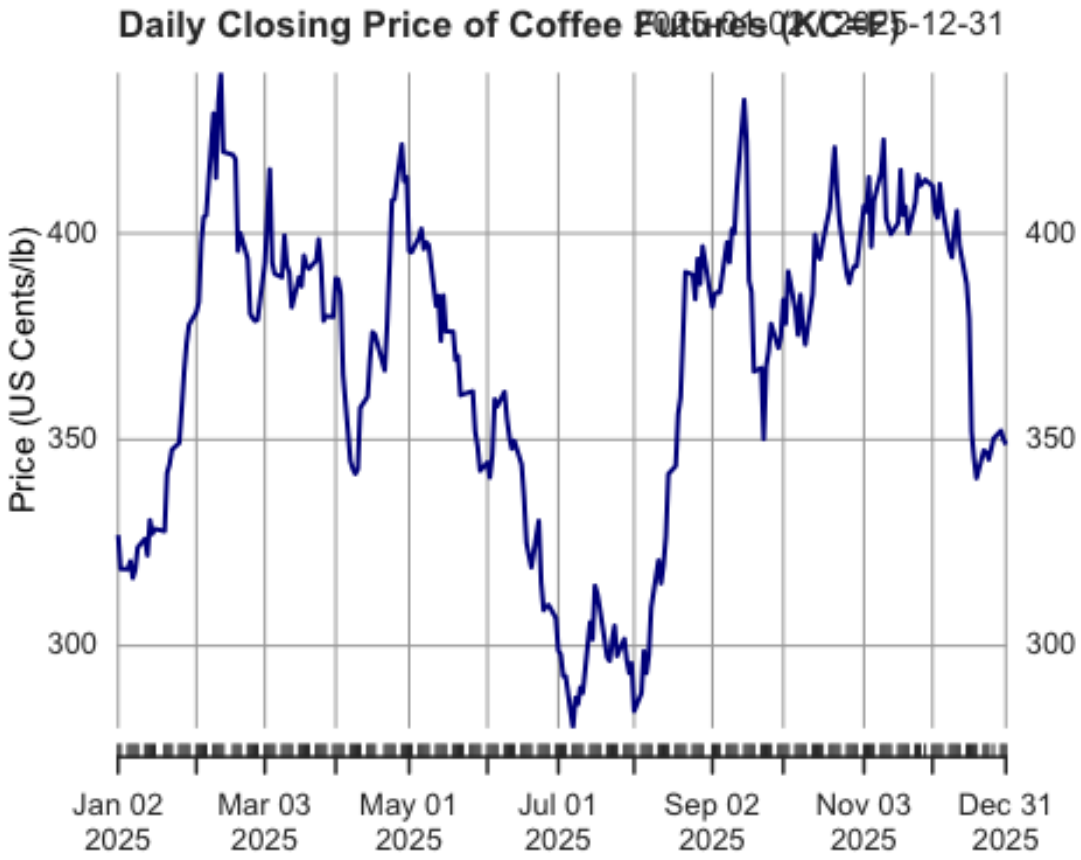
tail(`KC=F`)

##            KC=F.Open KC=F.High KC=F.Low KC=F.Close KC=F.Volume
KC=F.Adjusted
## 2025-12-23    350.00    352.40   345.60     346.95       11078
346.95
## 2025-12-24    348.10    348.65   343.05     345.15           0
345.15
## 2025-12-26    345.80    350.70   344.30     350.25        7348
350.25
## 2025-12-29    349.50    354.30   347.05     352.15        9184
352.15
## 2025-12-30    350.55    359.30   349.50     350.20       11564
350.20
```

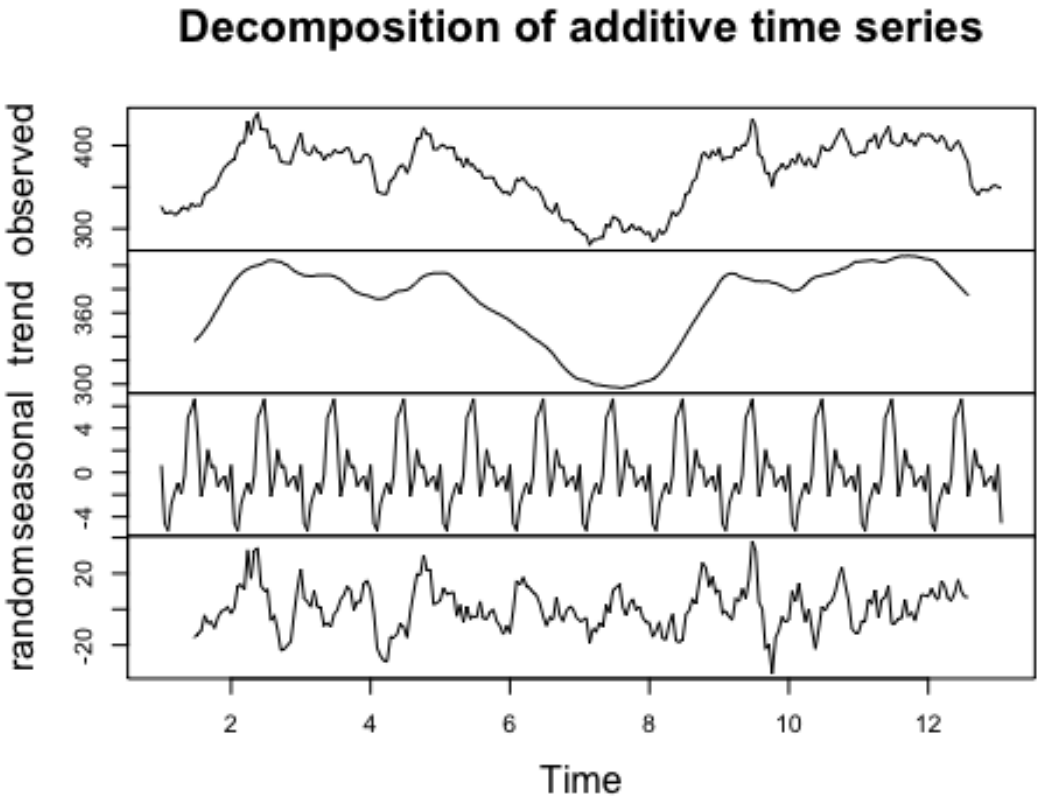
```
## 2025-12-31    349.50    353.85    348.05     348.75           0
348.75
```

```r
#extract closing price & clean NAs with linear interpolation
coffee_close <- `KC=F`[, "KC=F.Close"]
coffee_clean <- na.approx(coffee_close)

#plot the data
plot(coffee_clean, main="Daily Closing Price of Coffee Futures (KC=F)",
ylab="Price (US Cents/lb)", xlab="Date", col="darkblue", lwd=2)
```




```r
#ts decomposition
coffee_ts <- ts(as.numeric(coffee_clean), frequency=21)
coffee_decomp <- decompose(coffee_ts, type="additive")
plot(coffee_decomp)
```




```
#statistical verification
library(Kendall)
MannKendall(as.numeric(coffee_clean))

## tau = 0.0953, 2-sided pvalue =0.023706

months <- format(index(coffee_clean), "%m")
kruskal.test(as.numeric(coffee_clean) ~ months)

##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(coffee_clean) by months
## Kruskal-Wallis chi-squared = 165.06, df = 11, p-value < 2.2e-16

##partc
#split the data
train_coffee <- coffee_clean[1:(0.8*length(coffee_clean))]
train_coffee

##            KC.F.Close
## 2025-01-02     326.85
## 2025-01-03     318.65
## 2025-01-06     318.60
```

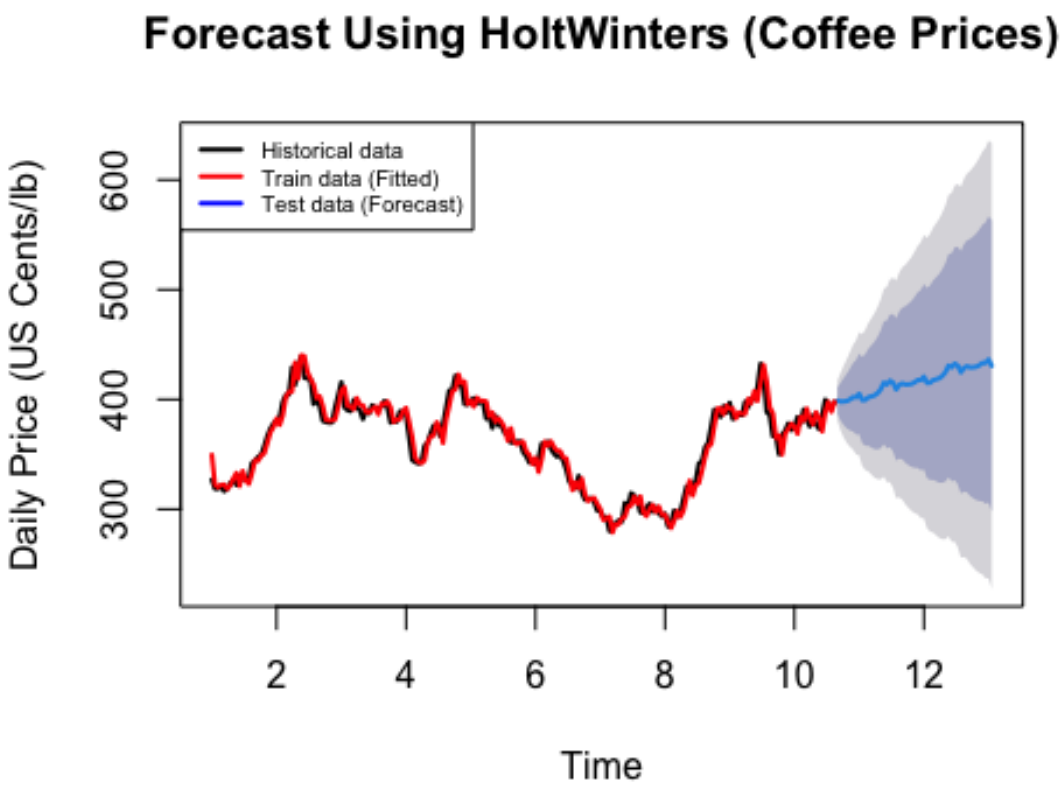
```
## 2025-01-07     320.50
## 2025-01-08     316.45
## 2025-01-09     318.50
## 2025-01-10     323.85
## 2025-01-13     325.95
## 2025-01-14     321.95
## 2025-01-15     330.45
##        ...
## 2025-10-06     381.35
## 2025-10-07     375.40
## 2025-10-08     385.10
## 2025-10-09     377.25
## 2025-10-10     373.05
## 2025-10-13     385.20
## 2025-10-14     399.65
## 2025-10-15     394.90
## 2025-10-16     393.80
## 2025-10-17     397.45
```

```
test_coffee <- coffee_clean[-c(1:(0.8*length(coffee_clean)))]
test_coffee
```

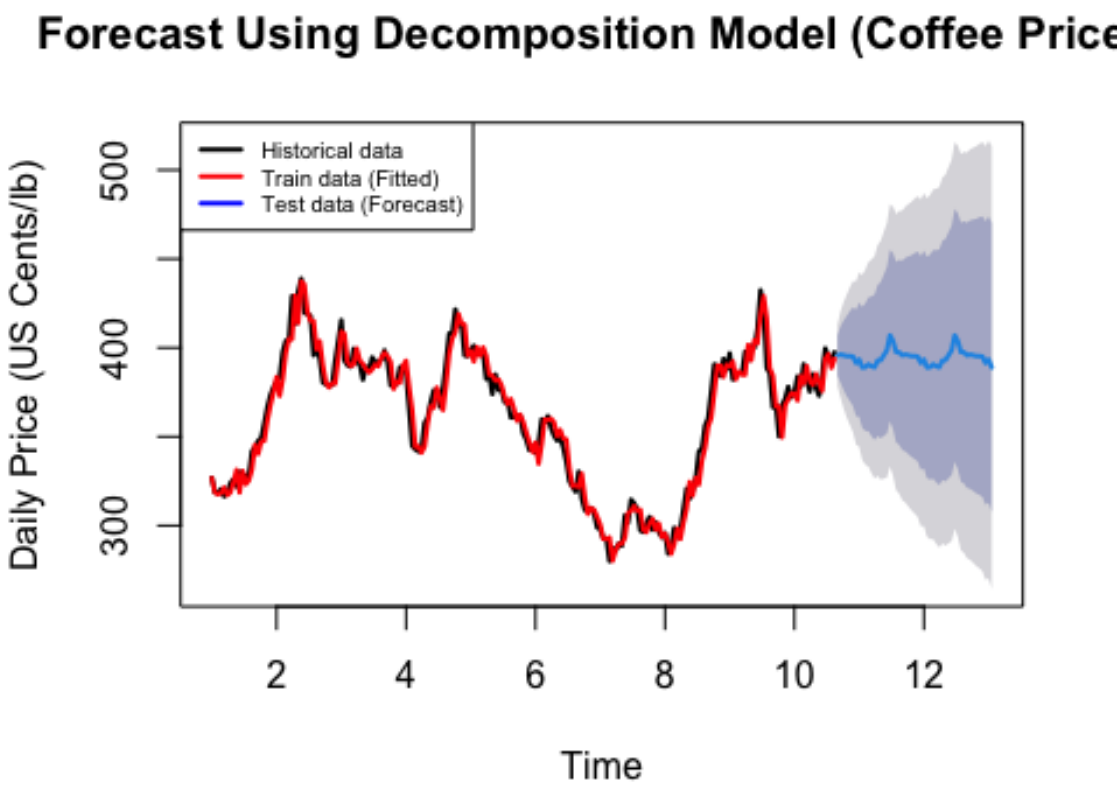
```
##            KC=F.Close
## 2025-10-20     406.05
## 2025-10-21     413.55
## 2025-10-22     420.85
## 2025-10-23     410.15
## 2025-10-24     403.00
## 2025-10-27     390.10
## 2025-10-28     387.90
## 2025-10-29     390.70
## 2025-10-30     392.00
## 2025-10-31     392.05
## 2025-11-03     406.65
## 2025-11-04     405.25
## 2025-11-05     413.60
## 2025-11-06     396.75
## 2025-11-07     407.80
## 2025-11-10     414.75
## 2025-11-11     422.70
## 2025-11-12     403.65
## 2025-11-13     401.70
## 2025-11-14     399.80
## 2025-11-17     402.55
## 2025-11-18     415.35
## 2025-11-19     404.50
## 2025-11-20     406.50
## 2025-11-21     400.00
## 2025-11-24     407.45
## 2025-11-25     414.20
```

```
## 2025-11-26     411.50
## 2025-11-28     413.00
## 2025-12-01     411.50
## 2025-12-02     405.55
## 2025-12-03     403.75
## 2025-12-04     411.95
## 2025-12-05     406.25
## 2025-12-08     396.05
## 2025-12-09     394.20
## 2025-12-10     400.80
## 2025-12-11     405.45
## 2025-12-12     397.20
## 2025-12-15     387.70
## 2025-12-16     379.30
## 2025-12-17     351.90
## 2025-12-18     345.45
## 2025-12-19     340.65
## 2025-12-22     347.35
## 2025-12-23     346.95
## 2025-12-24     345.15
## 2025-12-26     350.25
## 2025-12-29     352.15
## 2025-12-30     350.20
## 2025-12-31     348.75

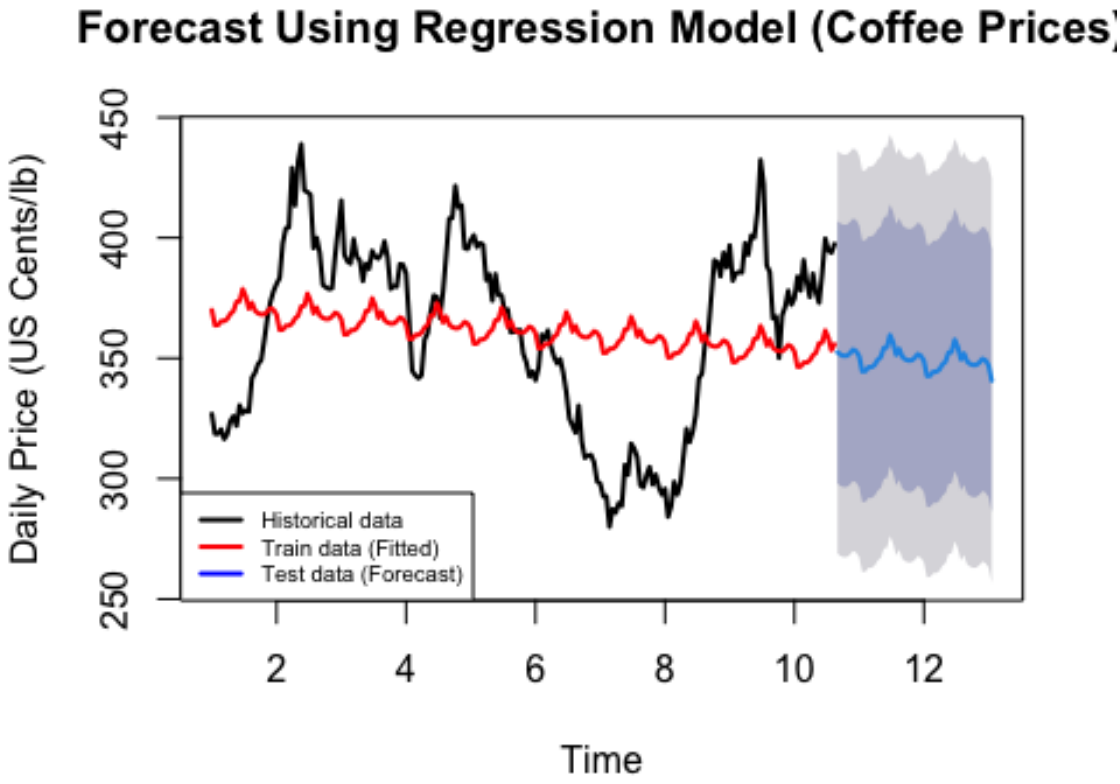
#add. holt-winters
train_coffee <- ts(as.numeric(train_coffee), frequency = 21)
h_days <- length(test_coffee)
library(forecast)
hw_forecast <- hw(train_coffee, seasonal="additive", h=h_days)
plot(hw_forecast, main="Forecast Using HoltWinters (Coffee Prices)",
ylab="Daily Price (US Cents/lb)", xlab="Time", lwd=2)
lines(hw_forecast$fitted, col="red", lwd=2)
legend("topleft", legend = c("Historical data", "Train data (Fitted)", "Test
data (Forecast)"), col = c("black", "red", "blue"), lty=1, lwd=2, cex=0.6)
```




```r
#add. decomp
decomp_forecast <- stlf(train_coffee, h=h_days)
plot(decomp_forecast, main="Forecast Using Decomposition Model (Coffee
Prices)", ylab="Daily Price (US Cents/lb)", xlab="Time", lwd=2)
lines(decomp_forecast$fitted, col="red", lwd=2)
legend("topleft", legend = c("Historical data", "Train data (Fitted)", "Test
data (Forecast)"), col = c("black", "red", "blue"), lty=1, lwd=2, cex=0.6)
```




```r
#reg
reg_model <- tslm(train_coffee ~ trend + season)
reg_forecast <- forecast(reg_model, h=h_days)
plot(reg_forecast, main="Forecast Using Regression Model (Coffee Prices)",
ylab="Daily Price (US Cents/lb)", xlab="Time", lwd=2)
lines(reg_forecast$fitted, col="red", lwd=2)
legend("bottomleft", legend = c("Historical data", "Train data (Fitted)",
"Test data (Forecast)"), col = c("black", "red", "blue"), lty=1, lwd=2,
cex=0.6)
```




```
##partd
accuracy(hw_forecast, test_coffee)
```

```
##                       ME      RMSE       MAE         MPE     MAPE     MASE
## Training set  -0.4106697  8.842443  6.898971 -0.1230875 1.903835 1.011095
## Test set     -23.3802853 39.867457 26.811417 -6.4923436 7.320524 3.929410
##                    ACF1
## Training set 0.01092945
## Test set             NA
```

```
accuracy(decomp_forecast, test_coffee)
```

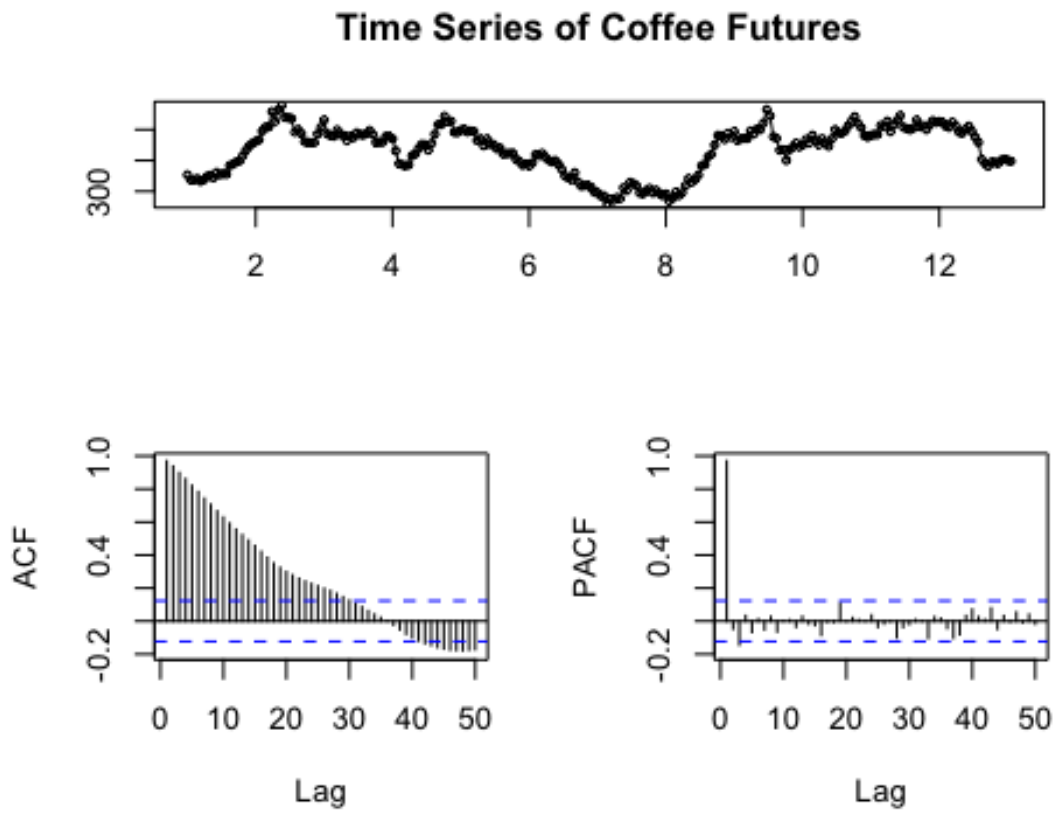
```
##                       ME      RMSE       MAE          MPE     MAPE      MASE
## Training set  0.3727976  8.179082  6.244091  0.07968001 1.716721 0.9151174
## Test set     -1.9092422 24.264322 18.728418 -0.88371193 4.982808 2.7447872
##                    ACF1
## Training set 0.01449186
## Test set             NA
```

```
accuracy(reg_forecast, test_coffee)
```

```
##                         ME     RMSE      MAE        MPE      MAPE     MASE
## Training set -1.071064e-14 37.66944 32.62141 -1.159628  9.375632 4.780908
## Test set      4.310941e+01 48.95950 43.64193 10.637964 10.793230 6.396045
```

```
##                     ACF1
## Training set 0.9671328
## Test set            NA
```

```
##parte
tsdisplay(coffee_ts, lag.max=50, main="Time Series of Coffee Futures")
```




```
#required library
library(tseries)
library(forecast)

#test for stationary
adf.test(coffee_ts)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  coffee_ts
## Dickey-Fuller = -2.3412, Lag order = 6, p-value = 0.4321
## alternative hypothesis: stationary
```

```
kpss.test(coffee_ts)
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  coffee_ts
## KPSS Level = 0.45391, Truncation lag parameter = 5, p-value = 0.05392
```

```
#differencing
ndiffs(coffee_ts)
```

```
## [1] 1
```

```
#test for stationary after applied 1st diff
adf.test(diff(coffee_ts))
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  diff(coffee_ts)
## Dickey-Fuller = -5.839, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss.test(diff(coffee_ts))
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  diff(coffee_ts)
## KPSS Level = 0.097502, Truncation lag parameter = 5, p-value = 0.1
```

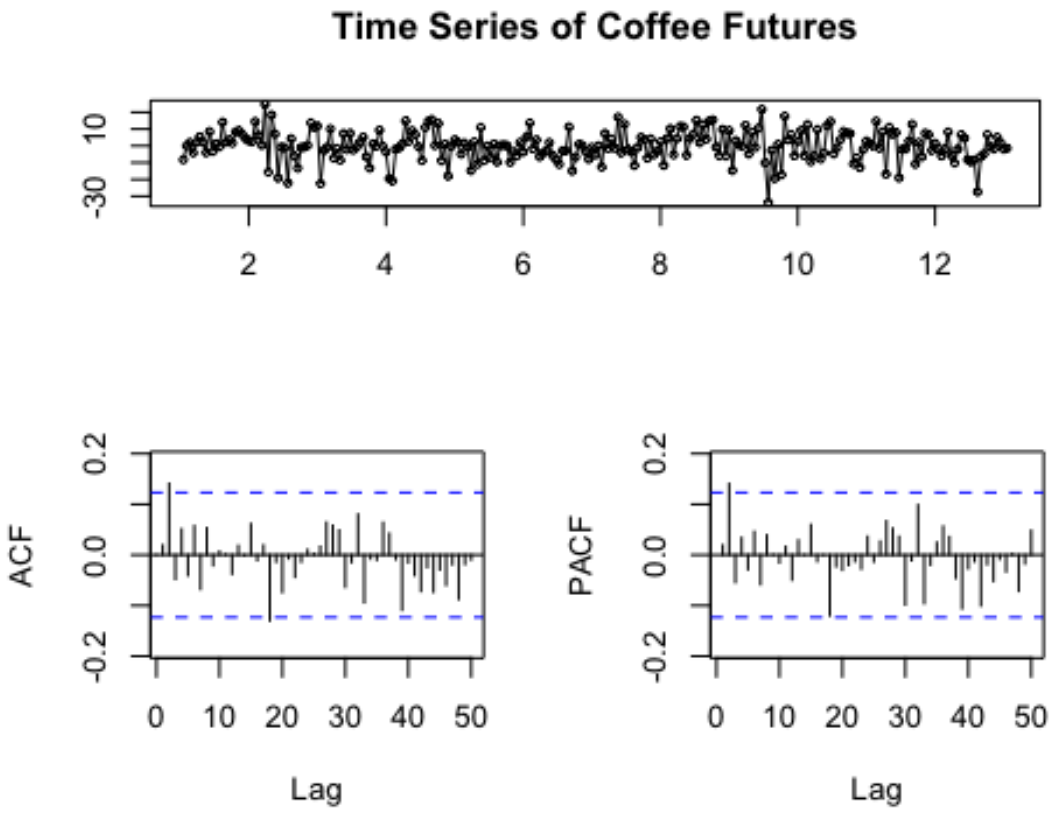
```
#test for seasonality
library(seastests)
isSeasonal(coffee_ts)
```

```
## [1] FALSE
```

```
nsdiffs(coffee_ts, test="ocsb")
```

```
## [1] 0
```

```
##partf
#fi
tsdisplay(coffee_ts, lag.max=50, main="Time Series of Coffee Futures")
tsdisplay(diff(coffee_ts), lag.max=50, main="Time Series of Coffee Futures")
```




```
auto.arima(coffee_ts, trace=TRUE)

##
##  Fitting models using approximations to speed things up...
##
##  ARIMA(2,1,2)(1,0,1)[21] with drift         : 1827.31
##  ARIMA(0,1,0)            with drift         : 1822.832
##  ARIMA(1,1,0)(1,0,0)[21] with drift         : 1841.171
##  ARIMA(0,1,1)(0,0,1)[21] with drift         : 1826.863
##  ARIMA(0,1,0)                               : 1820.824
##  ARIMA(0,1,0)(1,0,0)[21] with drift         : 1838.126
##  ARIMA(0,1,0)(0,0,1)[21] with drift         : 1824.864
##  ARIMA(0,1,0)(1,0,1)[21] with drift         : 1840.171
##  ARIMA(1,1,0)            with drift         : 1824.913
##  ARIMA(0,1,1)            with drift         : 1824.807
##  ARIMA(1,1,1)            with drift         : 1826.253
##
##  Now re-fitting the best model(s) without approximations...
##
##  ARIMA(0,1,0)                               : 1825.193
##
##  Best model: ARIMA(0,1,0)
```

```
## Series: coffee_ts
## ARIMA(0,1,0)
##
## sigma^2 = 78.91:  log likelihood = -911.59
## AIC=1825.18   AICc=1825.19   BIC=1828.71
```

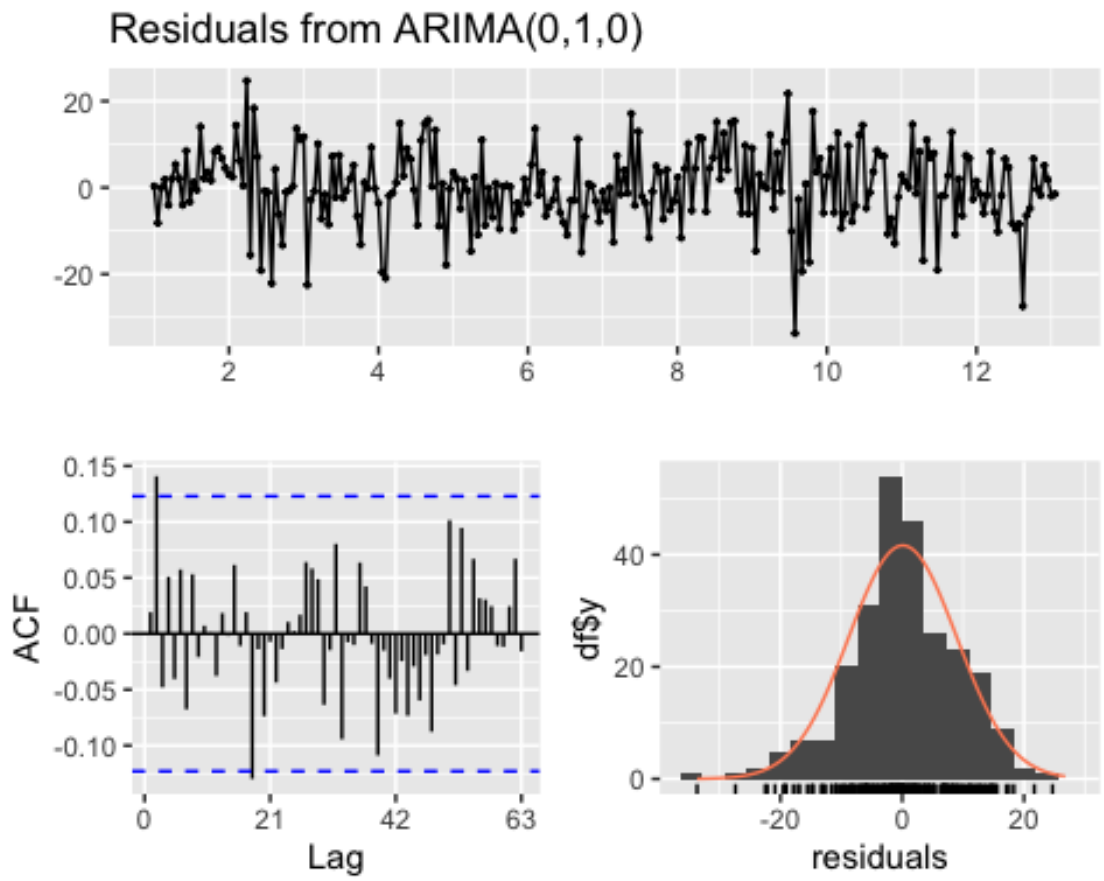
```
#fii
arima_coffee_010 <- Arima(coffee_ts, order=c(0,1,0))
arima_coffee_011 <- Arima(coffee_ts, order=c(0,1,1))
summary(arima_coffee_010)
```

```
## Series: coffee_ts
## ARIMA(0,1,0)
##
## sigma^2 = 78.91:  log likelihood = -911.59
## AIC=1825.18   AICc=1825.19   BIC=1828.71
##
## Training set error measures:
##                      ME     RMSE      MAE          MPE     MAPE    MASE
## Training set 0.08750726 8.865859 6.744201 -0.002296091 1.826752 0.18767
##                    ACF1
## Training set 0.01919458
```

```
summary(arima_coffee_011)
```

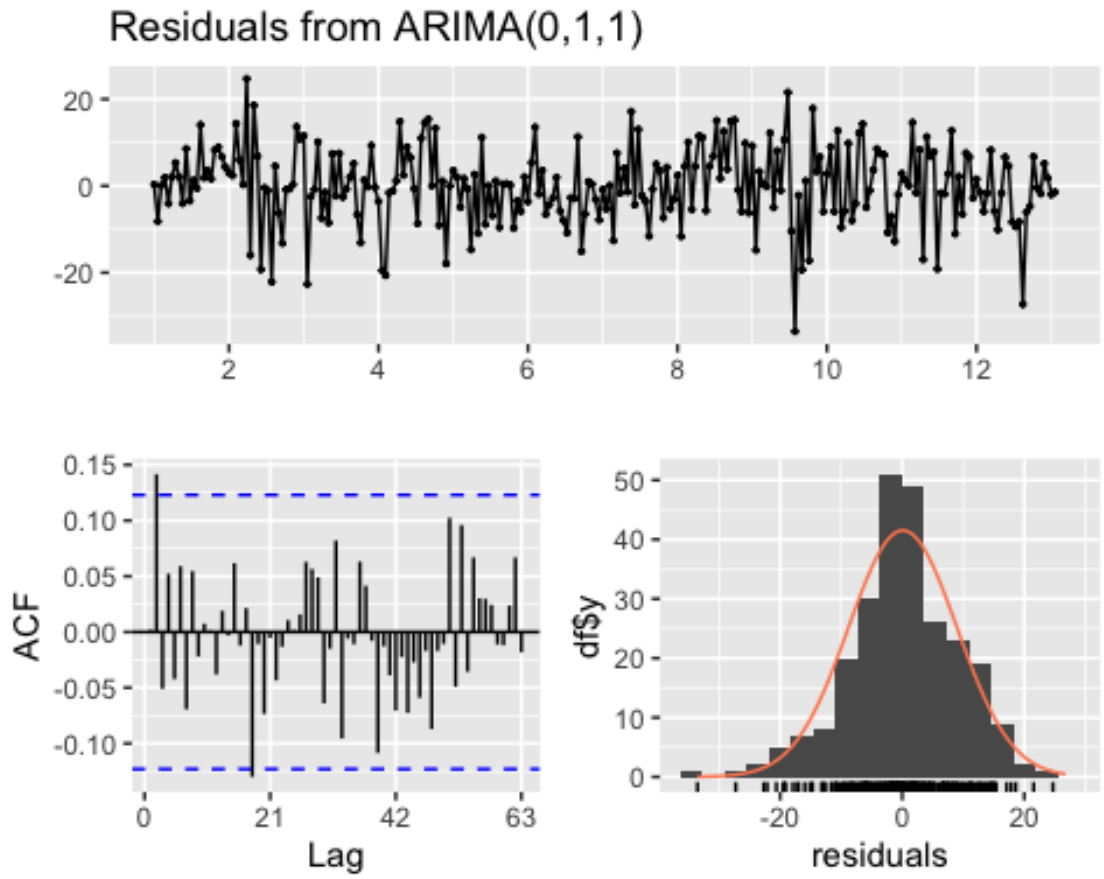
```
## Series: coffee_ts
## ARIMA(0,1,1)
##
## Coefficients:
##          ma1
##       0.0151
## s.e.  0.0554
##
## sigma^2 = 79.2:  log likelihood = -911.55
## AIC=1827.1   AICc=1827.15   BIC=1834.17
##
## Training set error measures:
##                      ME     RMSE      MAE          MPE     MAPE      MASE
## Training set 0.08614284 8.864552 6.742134 -0.002260039 1.826316 0.1876125
##                     ACF1
## Training set 0.001933793
```

```
#fiv
checkresiduals(arima_coffee_010)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,0)
## Q* = 34.534, df = 42, p-value = 0.7866
##
## Model df: 0.   Total lags used: 42
```

```
checkresiduals(arima_coffee_011)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,1)
## Q* = 34.74, df = 41, p-value = 0.7439
##
## Model df: 1.   Total lags used: 42
```

```
#fv
library(lmtest)
#coeftest(arima_coffee_010) is not included due to no coefficient to test
coeftest(arima_coffee_011)
```

```
##
## z test of coefficients:
##
##     Estimate Std. Error z value Pr(>|z|)
## ma1 0.015125   0.055410   0.273   0.7849
```

```
#fvi
summary(arima_coffee_010)
```

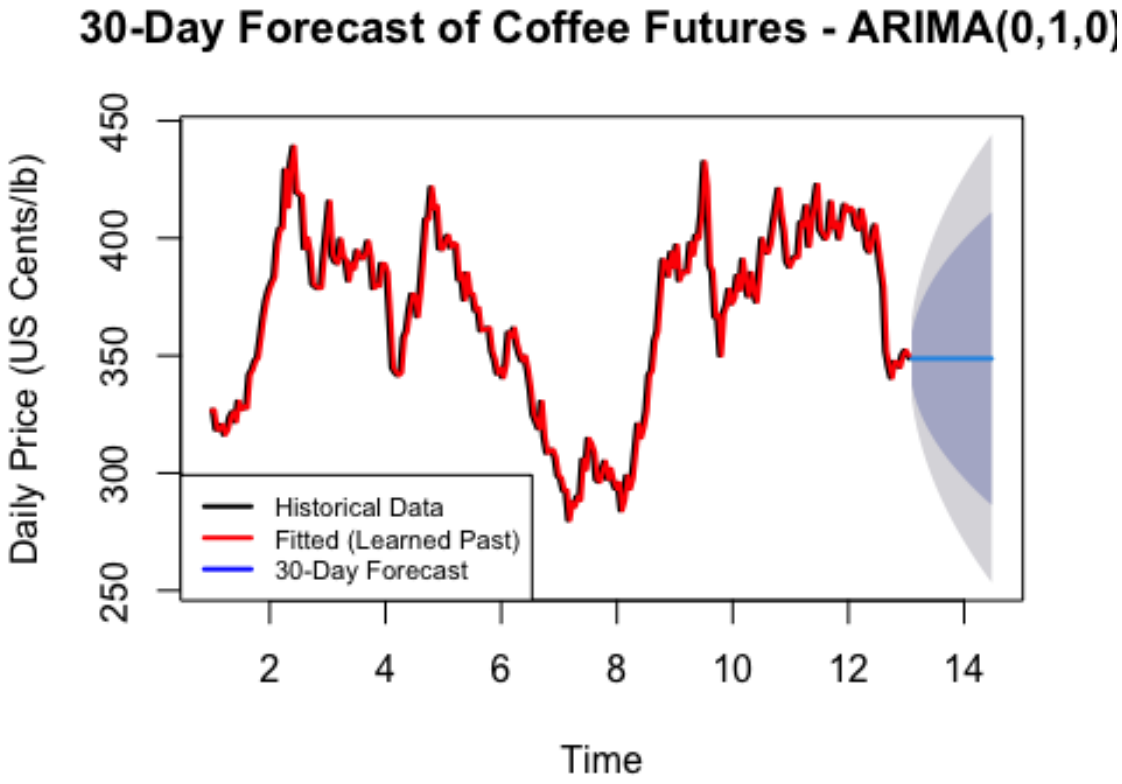
```
## Series: coffee_ts
## ARIMA(0,1,0)
```

```
##
## sigma^2 = 78.91:  log likelihood = -911.59
## AIC=1825.18   AICc=1825.19   BIC=1828.71
##
## Training set error measures:
##                       ME     RMSE      MAE          MPE     MAPE    MASE
## Training set 0.08750726 8.865859 6.744201 -0.002296091 1.826752 0.18767
##                     ACF1
## Training set 0.01919458
```

```
summary(arima_coffee_011)
```

```
## Series: coffee_ts
## ARIMA(0,1,1)
##
## Coefficients:
##          ma1
##       0.0151
## s.e.  0.0554
##
## sigma^2 = 79.2:  log likelihood = -911.55
## AIC=1827.1   AICc=1827.15   BIC=1834.17
##
## Training set error measures:
##                       ME     RMSE      MAE          MPE     MAPE      MASE
## Training set 0.08614284 8.864552 6.742134 -0.002260039 1.826316 0.1876125
##                     ACF1
## Training set 0.001933793
```

```
##partg
forecast_30_arima_010 <- forecast(arima_coffee_010, h=30)
plot(forecast_30_arima_010, main="30-Day Forecast of Coffee Futures -
ARIMA(0,1,0)", ylab="Daily Price (US Cents/lb)", xlab="Time", lwd=2)
lines(forecast_30_arima_010$fitted, col="red", lwd=2)
legend("bottomleft", legend=c("Historical Data", "Fitted (Learned Past)",
"30-Day Forecast"), col=c("black", "red", "blue"), lty=1, lwd=2, cex=0.7)
```


30-Day Forecast of Coffee Futures - ARIMA(0,1,0)
Daily Price (US Cents/lb)
250
300
350
400
450
2
4
6
8
10
12
14
Time
Historical Data
Fitted (Learned Past)
30-Day Forecast